

Lecture 8:-

Probability, Events, Scientific Discovery

Quantifying Chances

- Flip a coin:

↗ hard

Q: "What are the chances that it will land on its head?"

Not: "If I flip it 10 times, how many will be heads?"

↘ easy

There are two possibilities Head (H), Tail (T) $\rightarrow$ Events

0.5	0.5
H	T

$\rightarrow$ universe

$\Omega = \{H, T\}$

omega

probability

Chance of it being heads = $P(H) = 0.5$
 " " " " tails = $P(T) = 0.5$

The two axioms of Probability:-

- must lies in: $[0-1]$
- sum of all events must be 1

$P(H) = 1.2 \times$

$P(H) = 0.7, P(T) = 0.2$

Axioms are thing that you can't prove but everybody agrees on it because it make senses

H H
H T
T H
T T

$\rightarrow$ Flip it twice: $\{HH, HT, TH, TT\}$

The two flips are "independent".

$P(\{H, H\}) =$ "Both flips are heads."

0.25	0.25
HH	HT
0.25	0.25
TH	TT

$$P(\text{favorable event}) = \frac{\text{no. of way event can occur}}{\text{no. of all possible outcomes}}$$

$$= \frac{1}{4} = 0.25 = 25\%$$

"Any of the flips is a head." event

$A = \{HH, HT, TH\}$

$P(A) = 3/4 = 0.75$

Alternate way / Intuition:-

"Both flips are heads"
 First flip is H and second flip is H
 So, $P(H) * P(H)$
 $= 0.5 * 0.5 = 0.25$ ✓

"Any of the flips is a head"
 First flip is H or second flip is H
 $P(H) + P(H)$
 $= 0.5 + 0.5$
 $= 1$ ✗

So

$$P(H) + P(H) - P(HH)$$

$$0.5 + 0.5 - 0.25 = 0.75$$
 ✓

Why? Error?
 Due to overlap
 or count two times
 of HH.

AND		
H0	H0	0
H0	T1	0
T1	H0	0
T1	T1	1

This is overlap

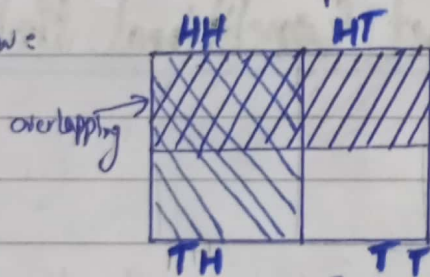
OR		
H0	H0	0
H0	T1	1
T1	H0	1
T1	T1	1

To coin one there occurs 2 time head same with coin 2 but we are count and event two time

remove overlap by subtracting its probability
 $= P(HH) = 0.25$

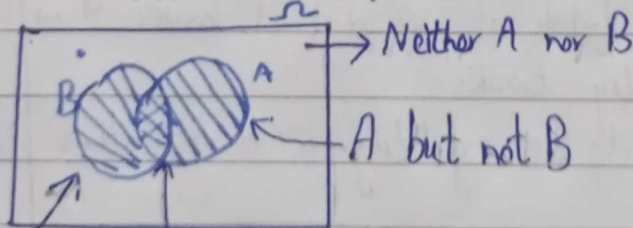
A and B are not independent!

Another view:



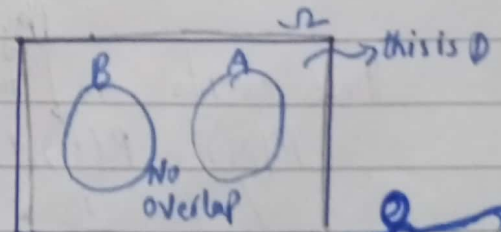
First flip H A
 Second flip H B

or



B but not A
 A and B

Single flip - independent events



$$P(A \text{ or } B) = P(A) + P(B) + P(A \text{ and } B)$$

Notation:

$$A \text{ and } B = A \cap B$$

$$A \text{ or } B = A \cup B$$

Let's scale this up to 1 million flips!
In Jupyter Notebook!